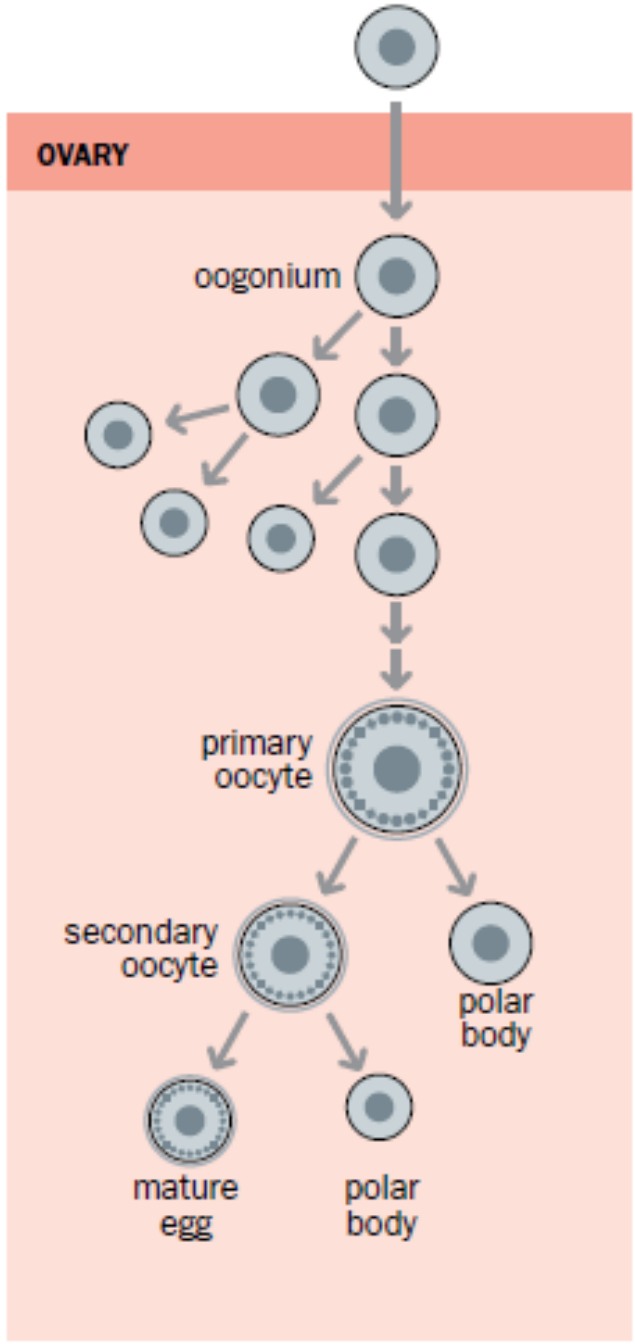
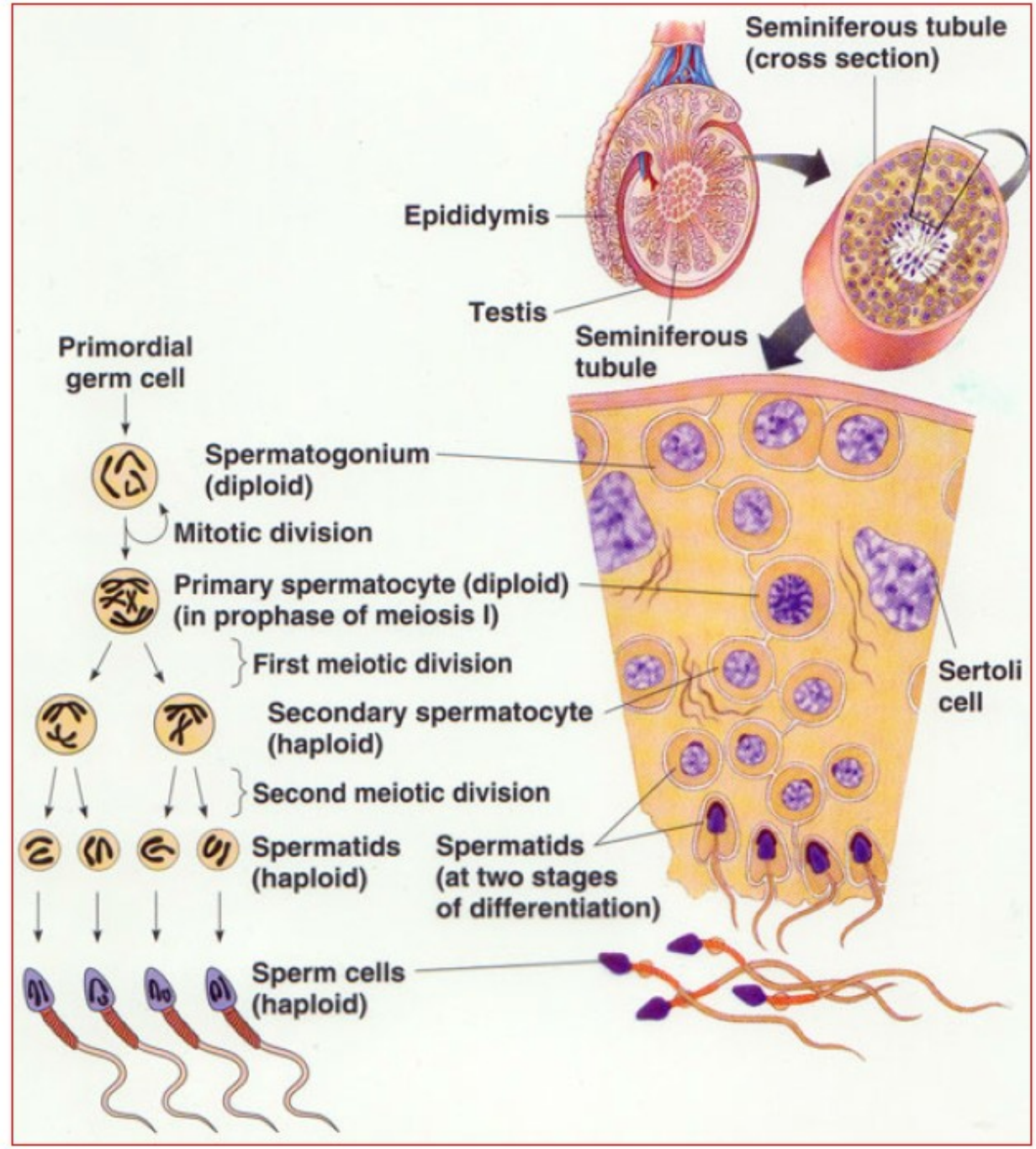
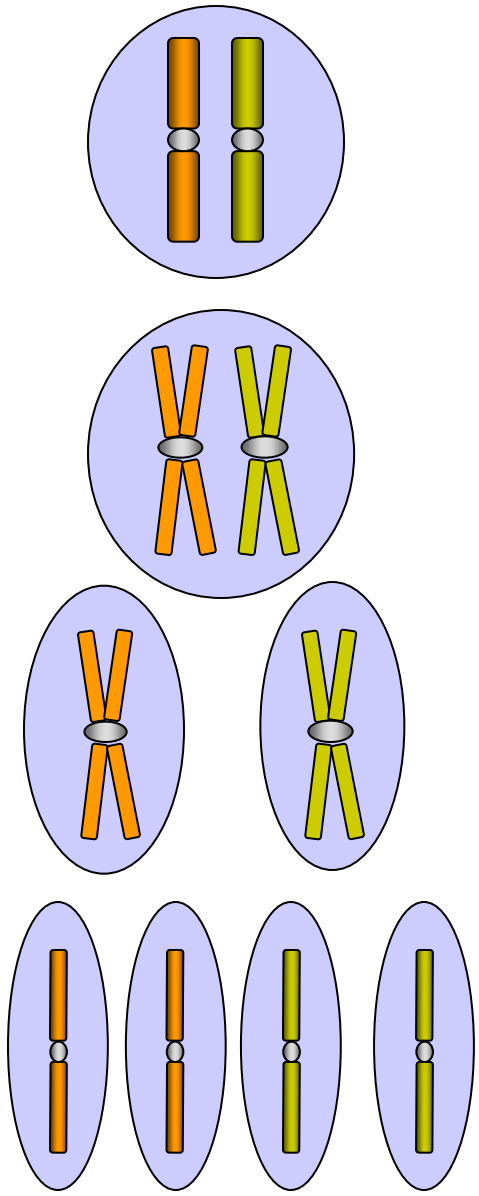
Lec 7.2: MEIOSIS

Learning goals:

- Formation of gametes requires a special type of cell division
- Meiosis includes two segregation steps (and two cell divisions)
- In order to increase genetic diversity in the offspring, meiosis includes HOMOLOGOUS RECOMBINATION by which homologous chromosomes exchange segments of DNA


OVARY
oogonium
primary oocyte
secondary oocyte
polar body
mature egg
polar body


Seminiferous tubule (cross section)
Epididymis
Testis
Seminiferous tubule
Primordial germ cell
Spermatogonium (diploid)
Mitotic division
Primary spermatocyte (diploid) (in prophase of meiosis I)
First meiotic division
Secondary spermatocyte (haploid)
Second meiotic division
Spermatids (haploid)
Spermatids (at two stages of differentiation)
Sertoli cell
Sperm cells (haploid)



Gonium (2n/2C)

Replication (S phase)

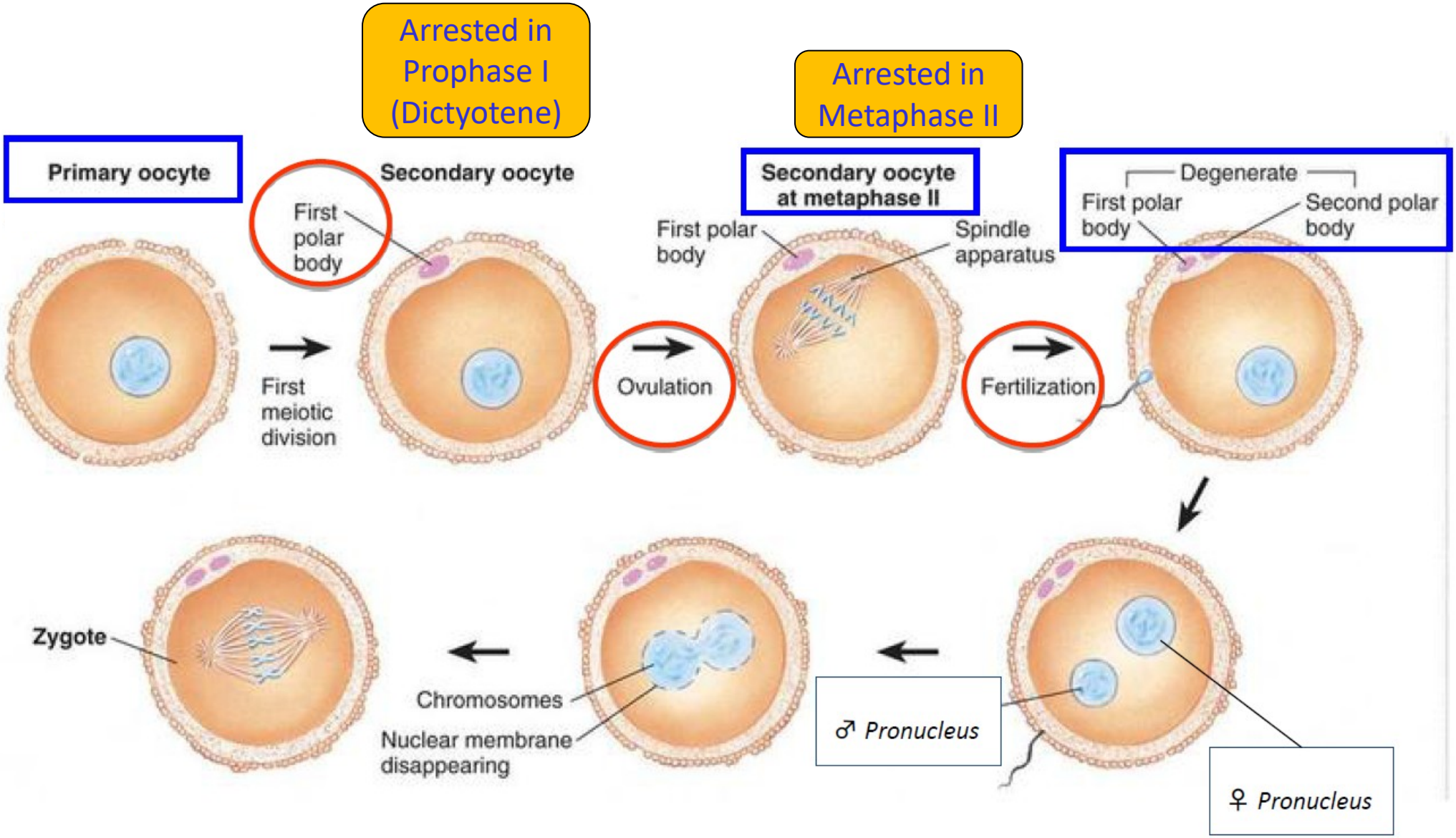
Primary Oocyte/Spermatocyte (2n/4C)

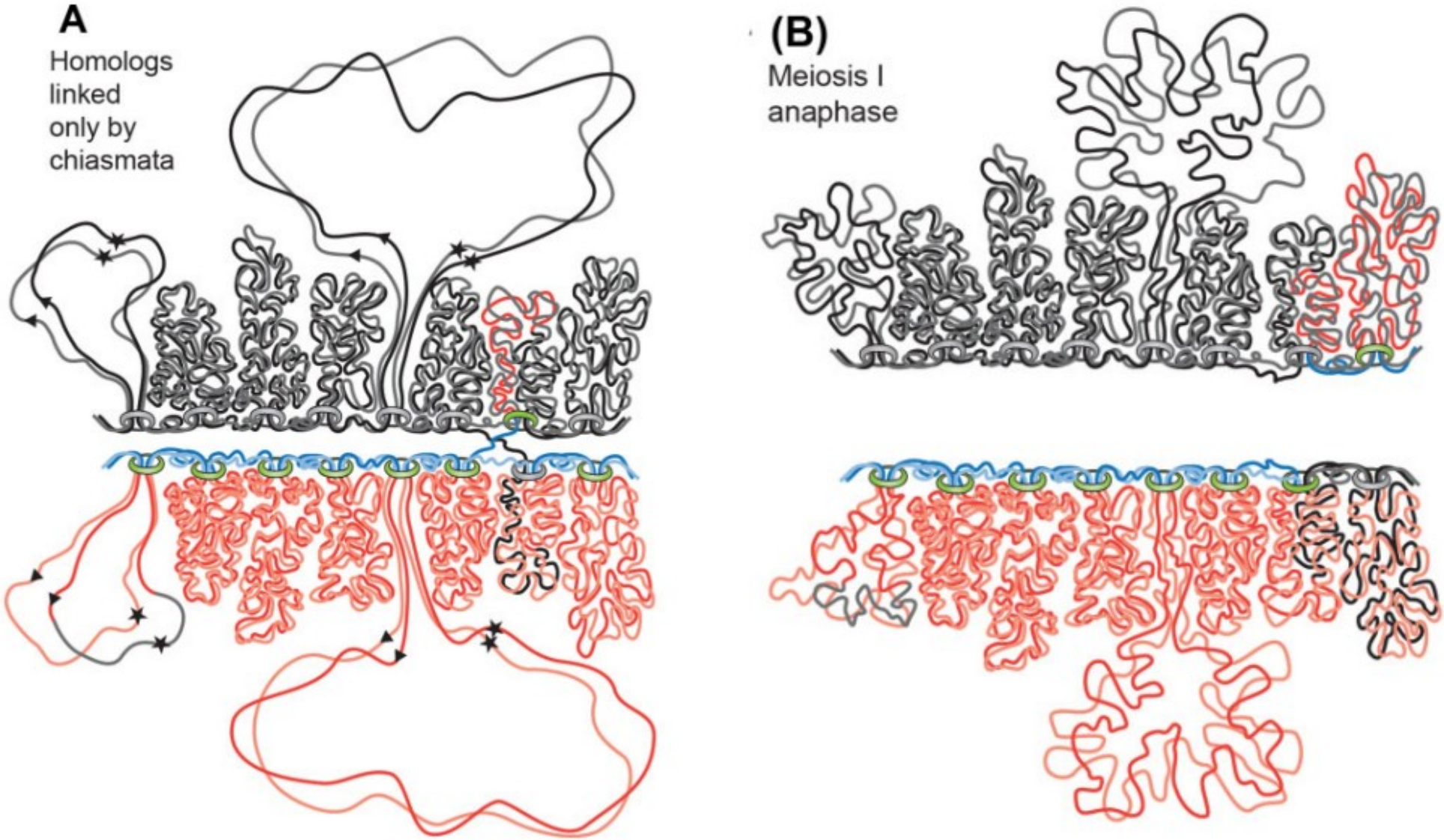
Meiosis I

Secondary Oocyte/Spermatocyte (n/2C)

Meiosis II

Gamete (n/1C)


Arrested in Prophase I (Dictyotene)
Arrested in Metaphase II
Primary oocyte
First polar body
Secondary oocyte
First meiotic division
Ovulation
Secondary oocyte at metaphase II
First polar body
Spindle apparatus
Fertilization
Degenerate
First polar body
Second polar body
Zygote
Chromosomes
Nuclear membrane disappearing
♂ Pronucleus
♀ Pronucleus

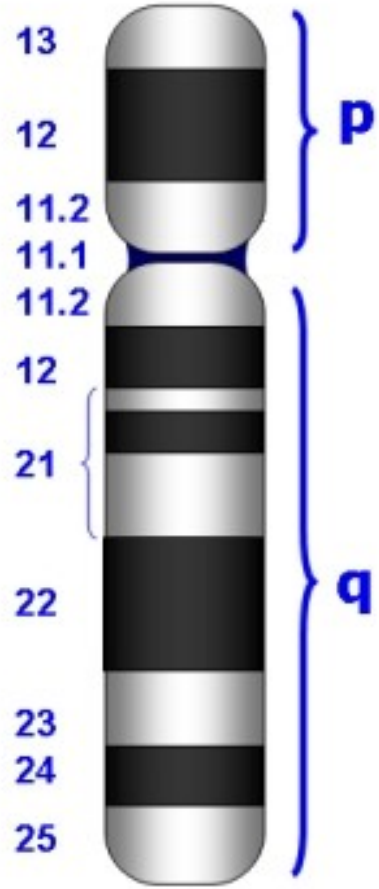

A
Homologs
linked
only by
chiasmata
(B)
Meiosis I
anaphase

Lec 7.4: LEVELS OF CHROMATIN PACKAGING

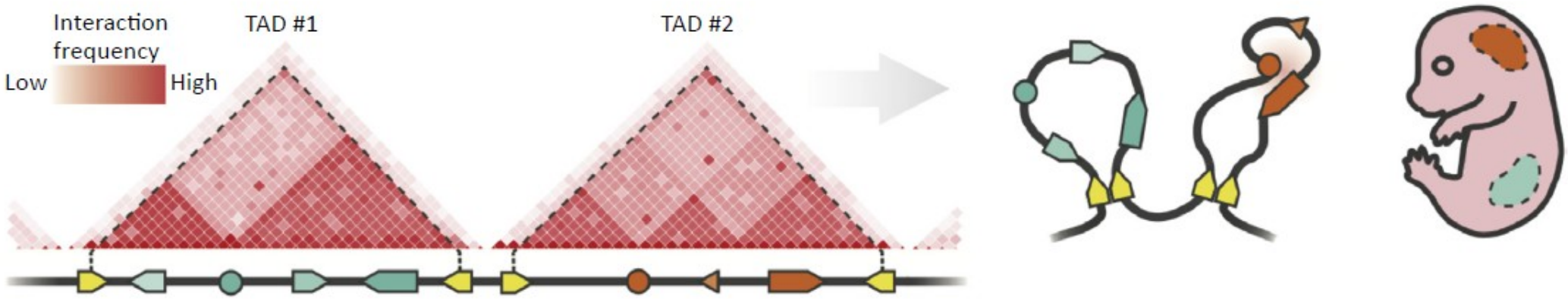
Learning goals:

- DNA in the linear genome is packaged into chromatin, and chromatin into chromosomes
- Chromosomes occupy territories in the nucleus
- Loops and compartments (formed at different levels of resolution) can be active or inactive

Metaphase chromosome 17





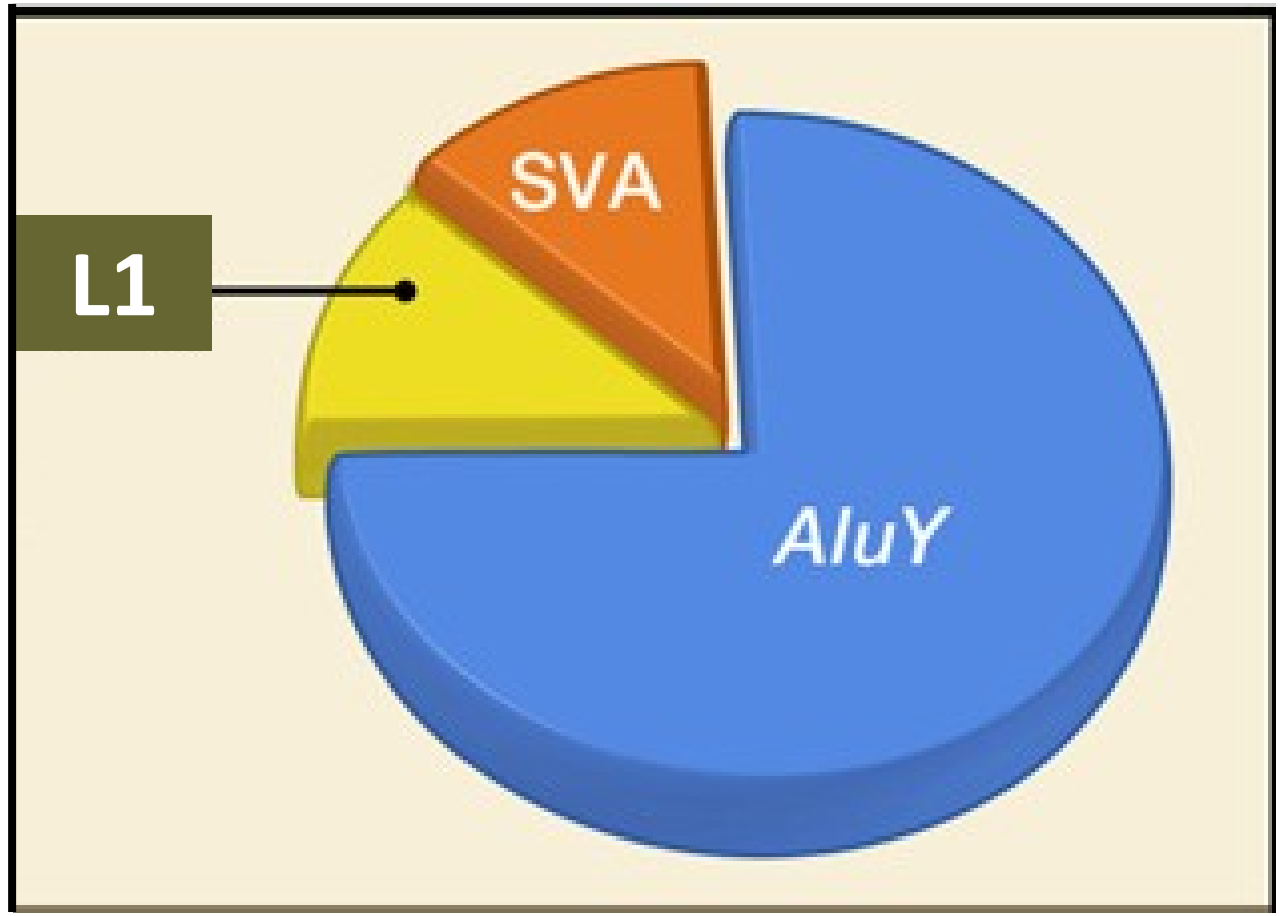

Interaction
frequency
Low
High
TAD #1
TAD #2

Lec 7.5: GENERAL FEATURES OF THE HUMAN GENOME

Learning goals:

- The human genome is 3Gb long and contains thousands of genes
- The content of G+C correlates with functional regions of the genome
- The genome is full of repeated copies of mobile elements

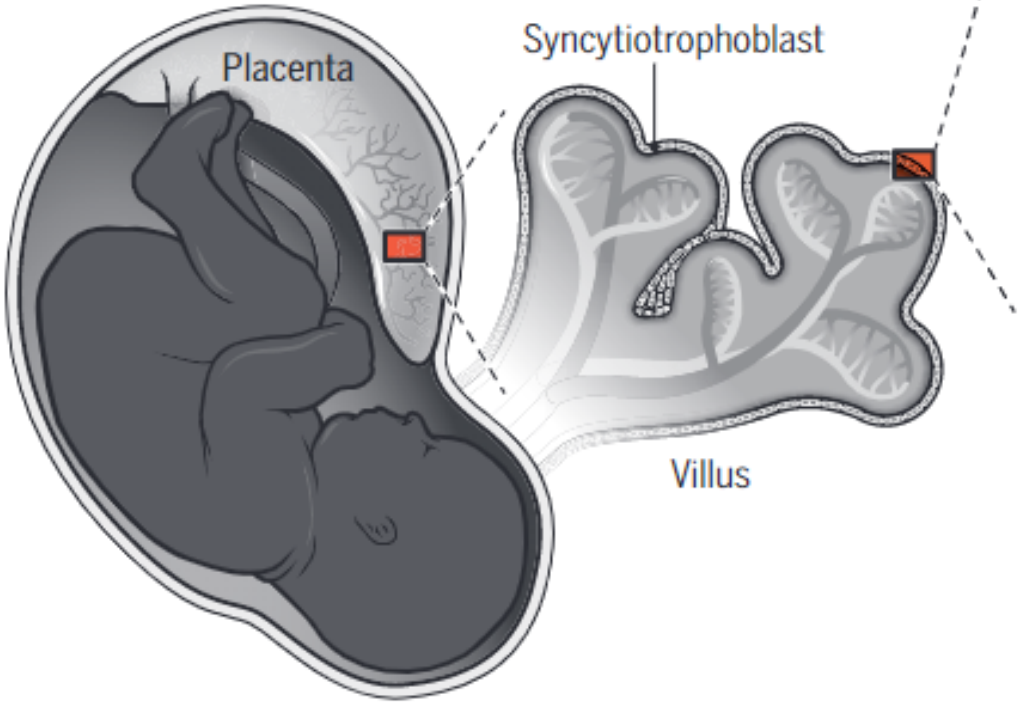
MAIN ACTIVE TRANSPOSABLE ELEMENTS



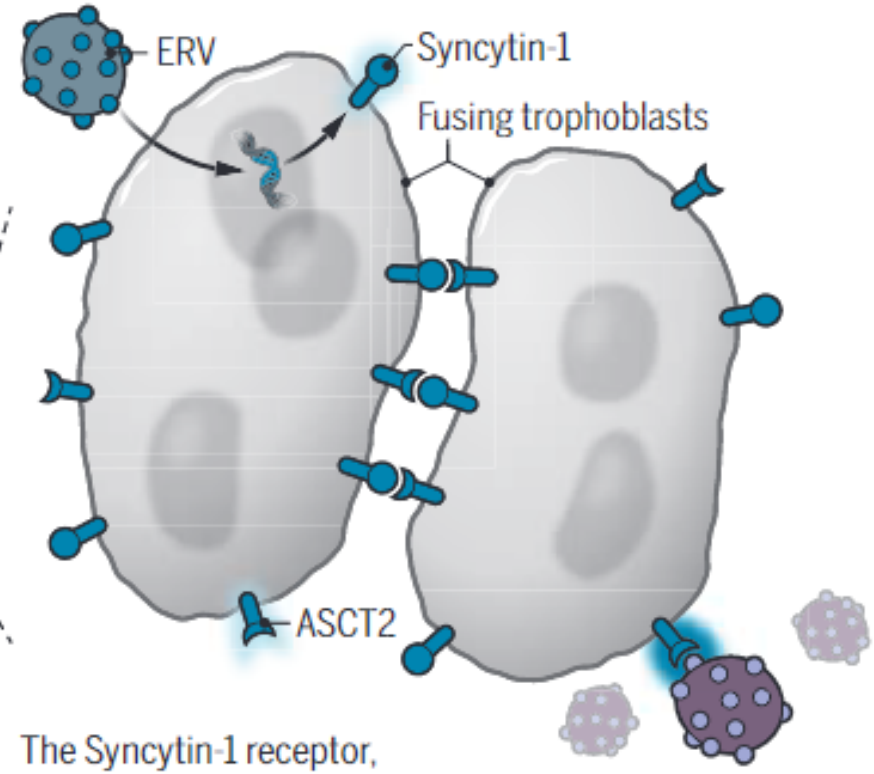
One *AluY* new germ-line insertion per 20 births

Domestication of endogenous retrovirus genes

The placental syncytiotrophoblast is an important barrier between maternal and fetal blood that is formed by cell-cell fusion. The fusogenic property of trophoblasts derives from domestication of ERV genes, but this came at a cost that prompted further selection for ERV gene domestication.



Domestication of an ASCT2-tropic ERV brings about expression of **Syncytin proteins.**



The Syncytin-1 receptor, ASCT2, is also expressed. Ligand-receptor binding **allows cell-cell fusion** and syncytia formation.

But this also leaves cells **susceptible** to other ASCT2-tropic ERVs.

ASCT2, alanine, serine, cysteine-preferring transporter 2; ERV, endogenous retrovirus.

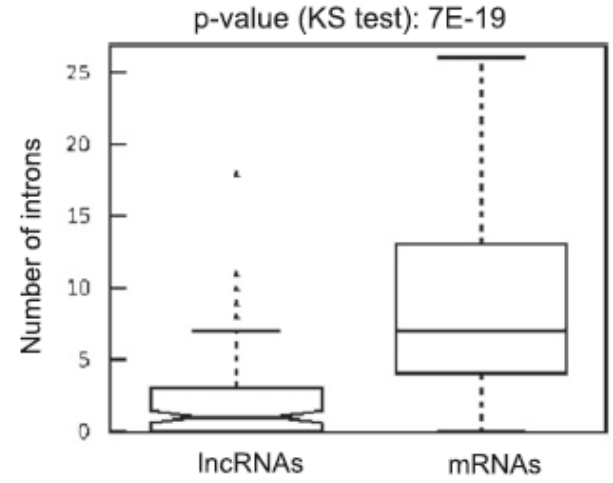
Lec 7.6: GENES (protein-coding vs. non-coding)

Learning goals:

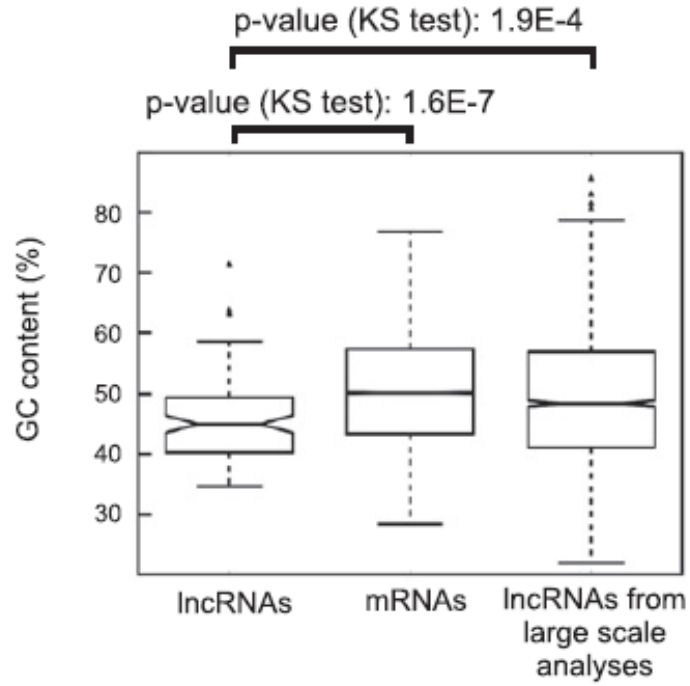
- Genes come in two classes: protein-coding or non-coding
- Abundance, size, characteristics and function of human genes

lncRNA genes are shorter than protein-coding

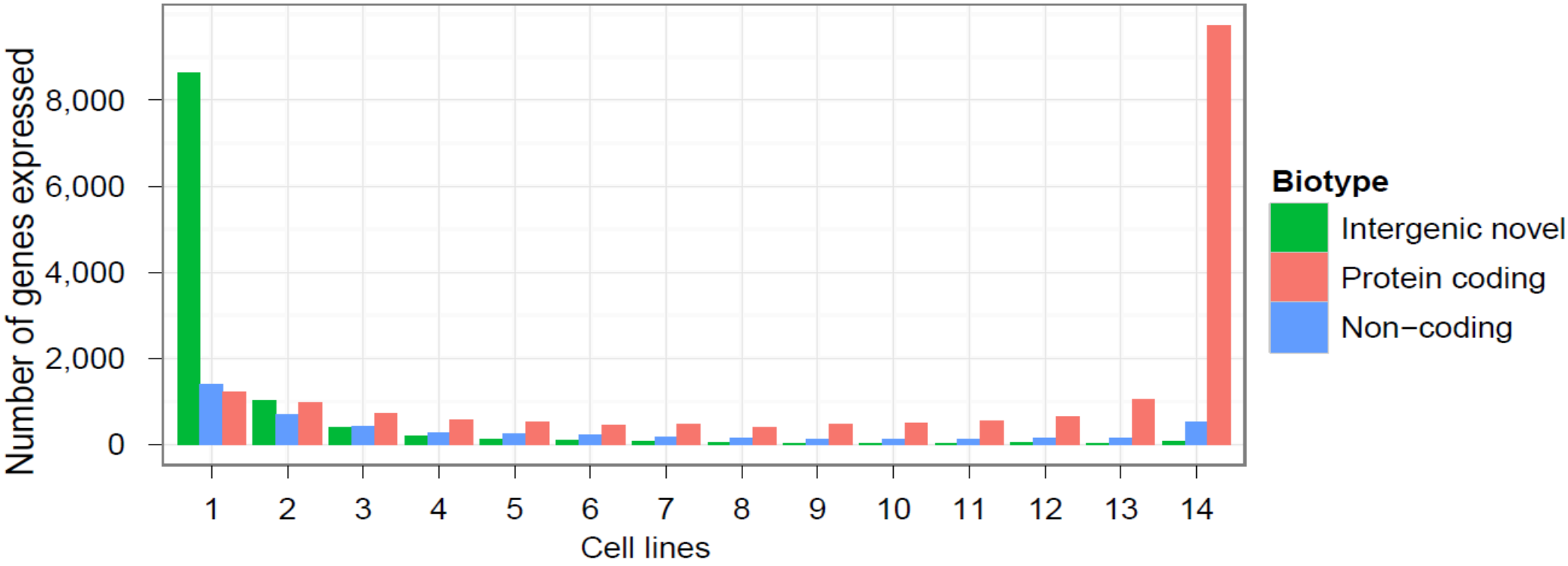
lncRNA genes have fewer exons



lncRNA genes are **NOT** GC-rich



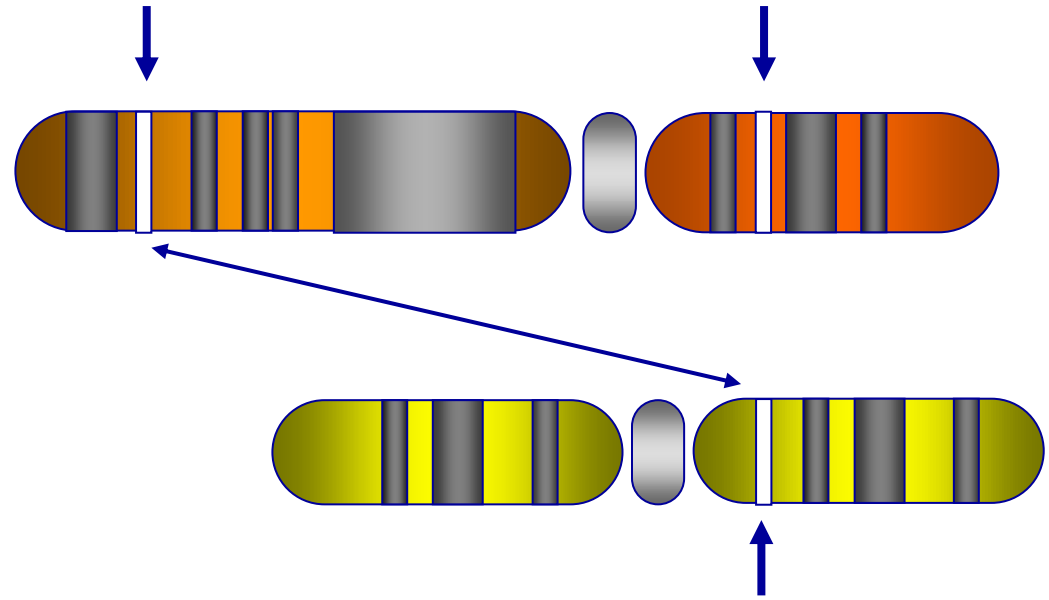
Expression of lncRNAs is **very resctricted**: tissue-specific differences



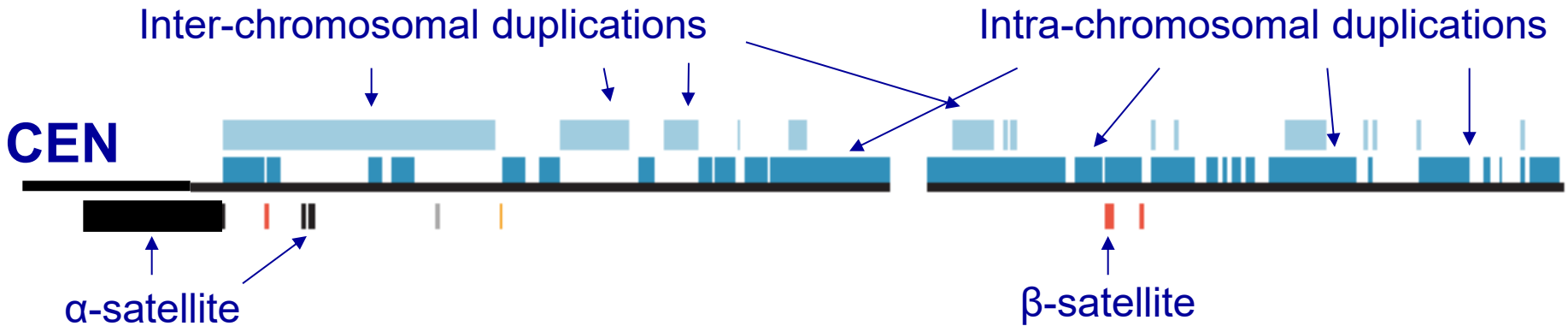
Protein coding: 7% expressed only in one cell line
50% expressed in all cell lines

lncRNAs: 30% expressed only in one cell line
10% expressed in all cell lines

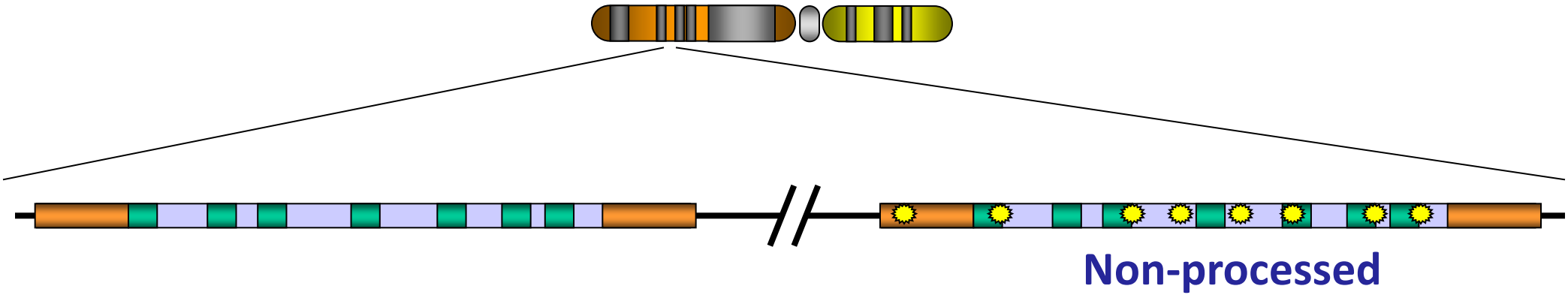
SEGMENTAL DUPLICATIONS



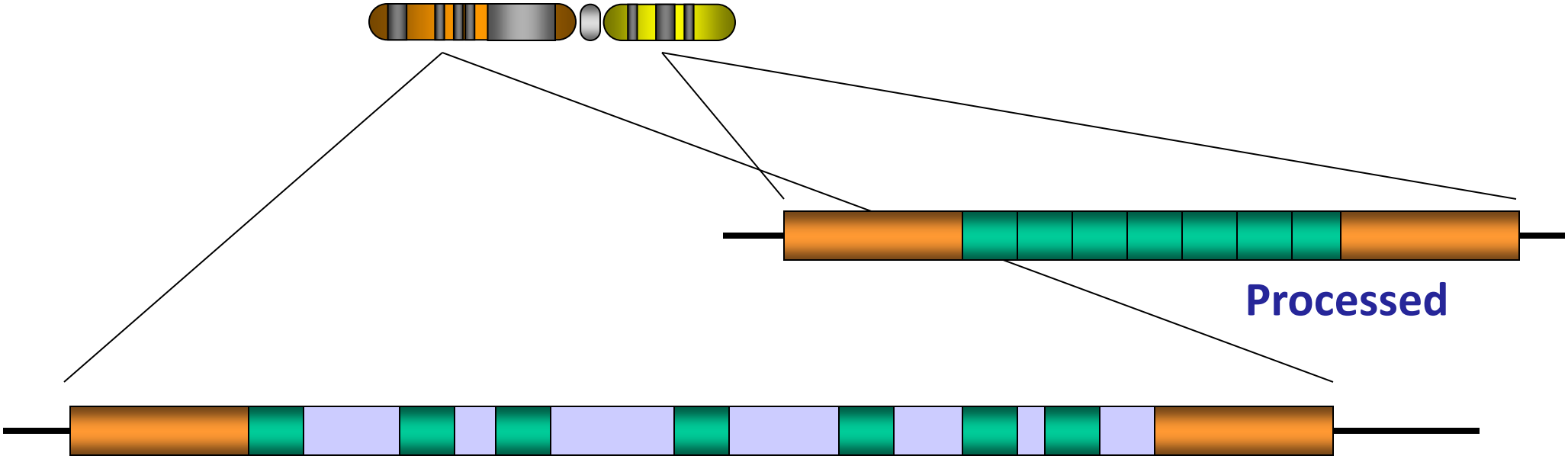
Chromosome 7p



PSEUDOGENES



Non-processed



Processed

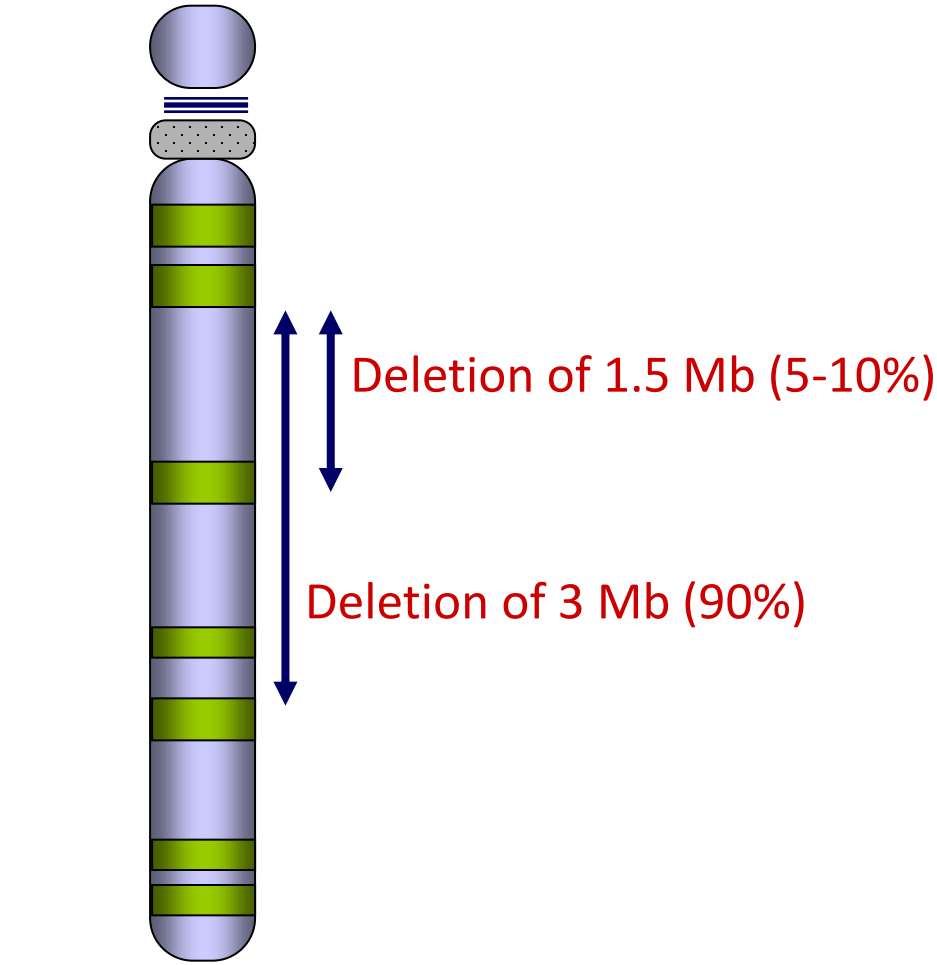
Lec 7.7: GENETIC VARIATION

Learning goals:

- Individual genomes differ in single-nucleotide polymorphisms and in larger structural variants
- These variants have profound implications for predisposition to disease

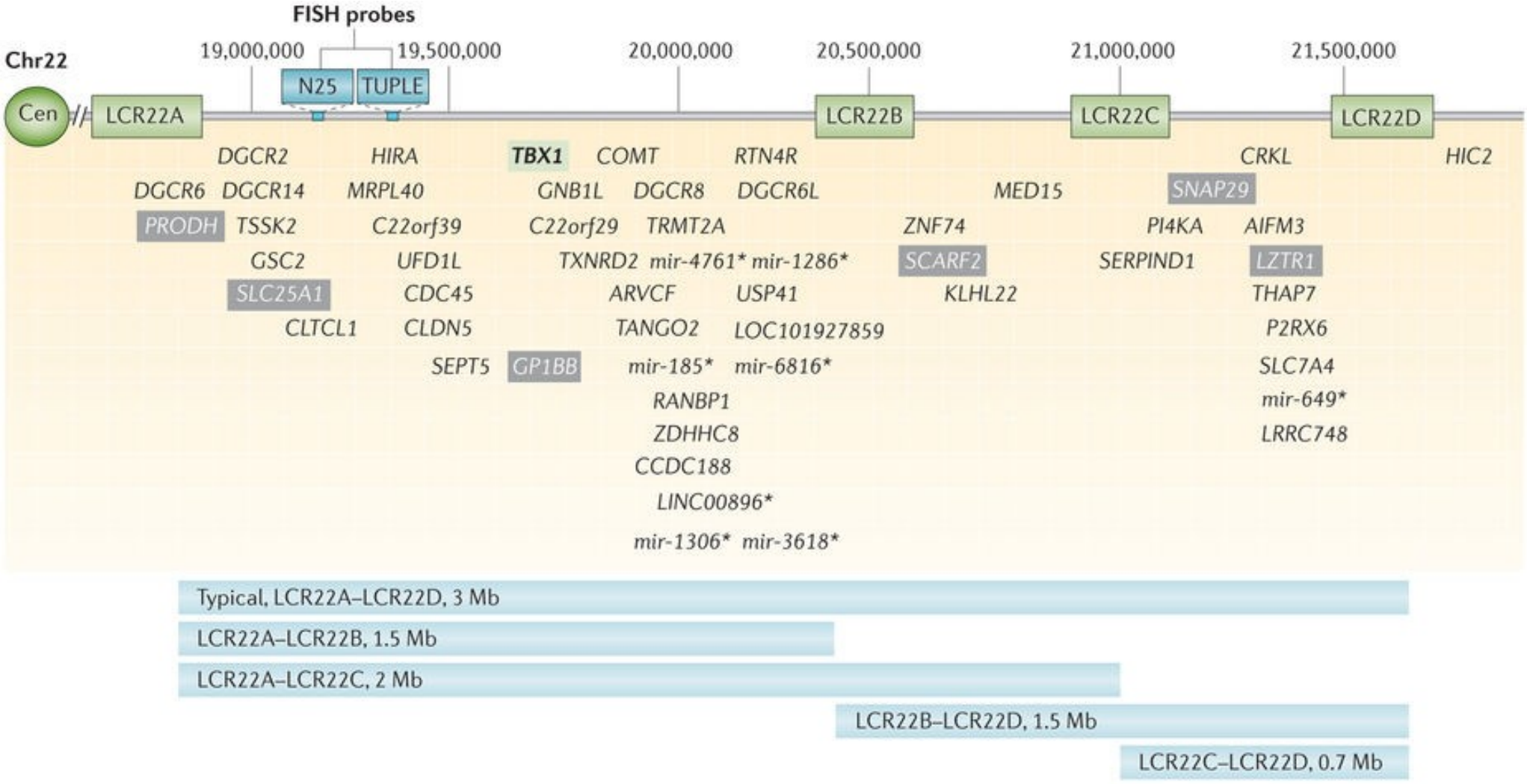
Genomic Disorders

(microdeletion syndromes, duplications and inversions)



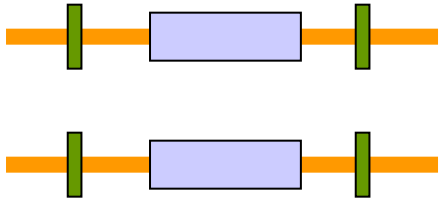
Di George (22q)

Di George (22q11DS)

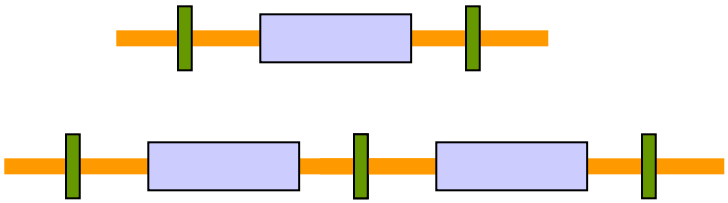


Nature Reviews | Disease Primers

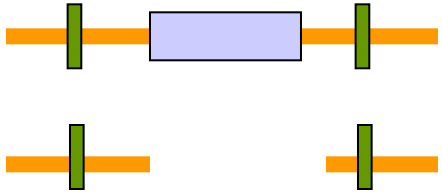
NORMAL

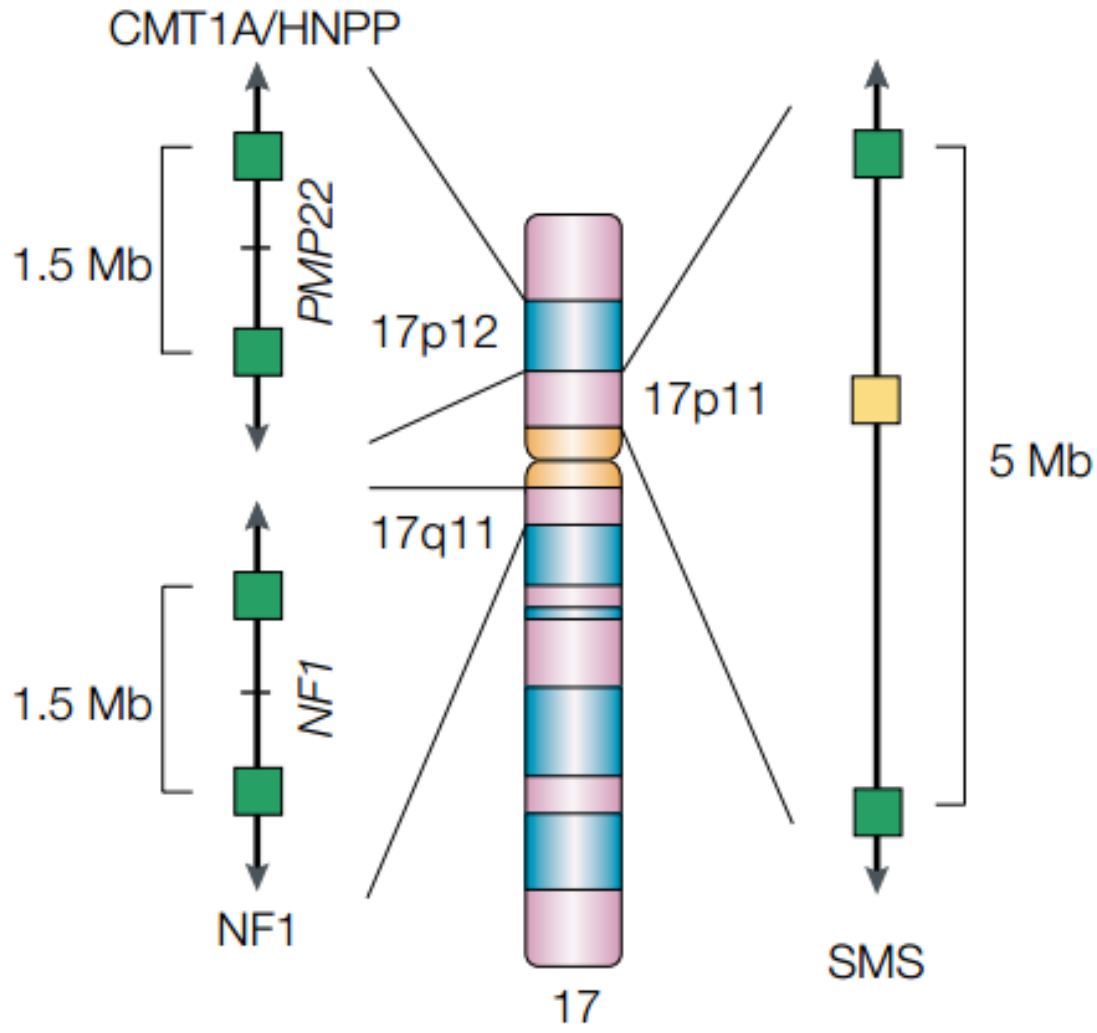


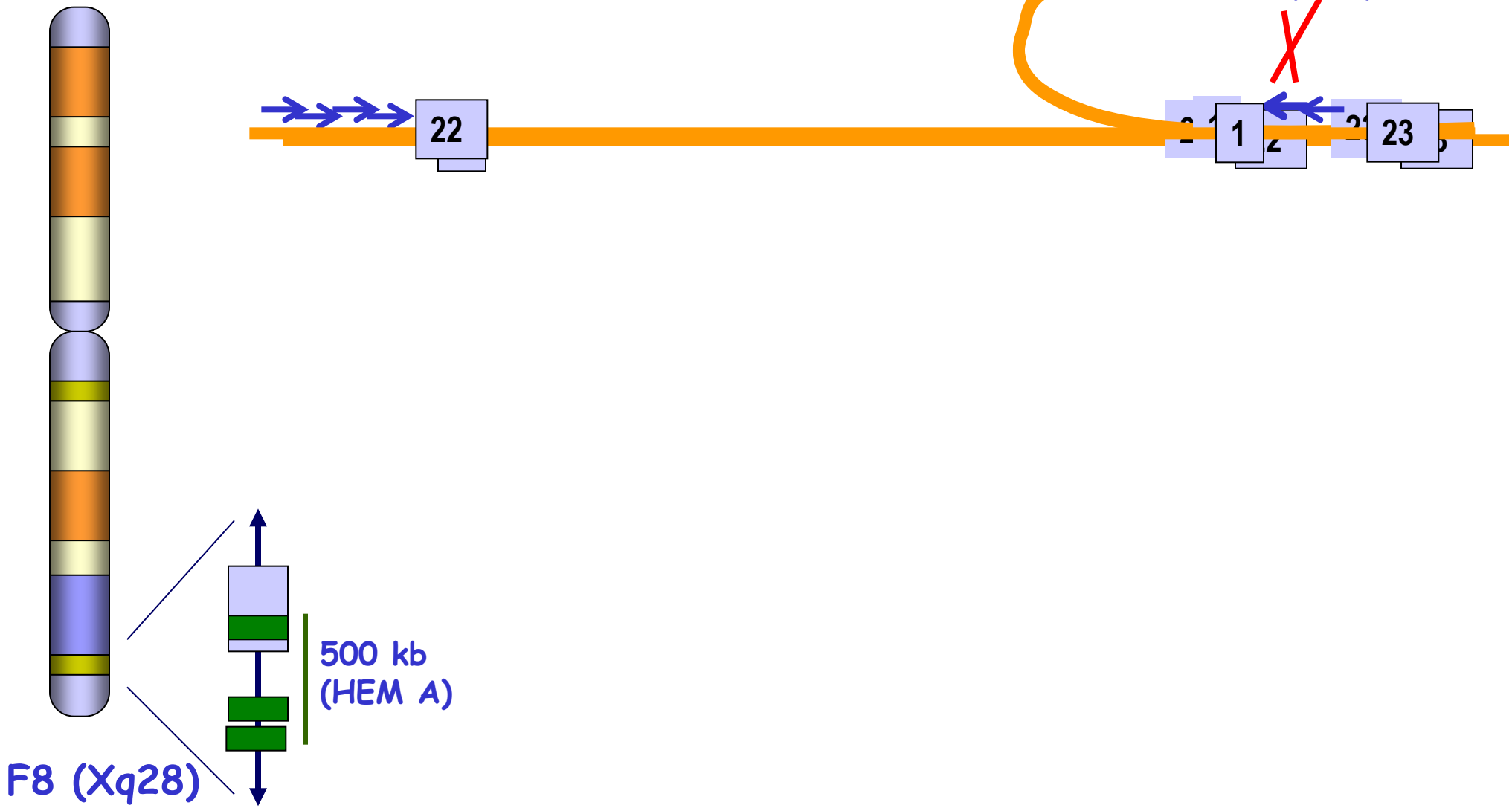
CHARCOT-MARIE-TOOTH



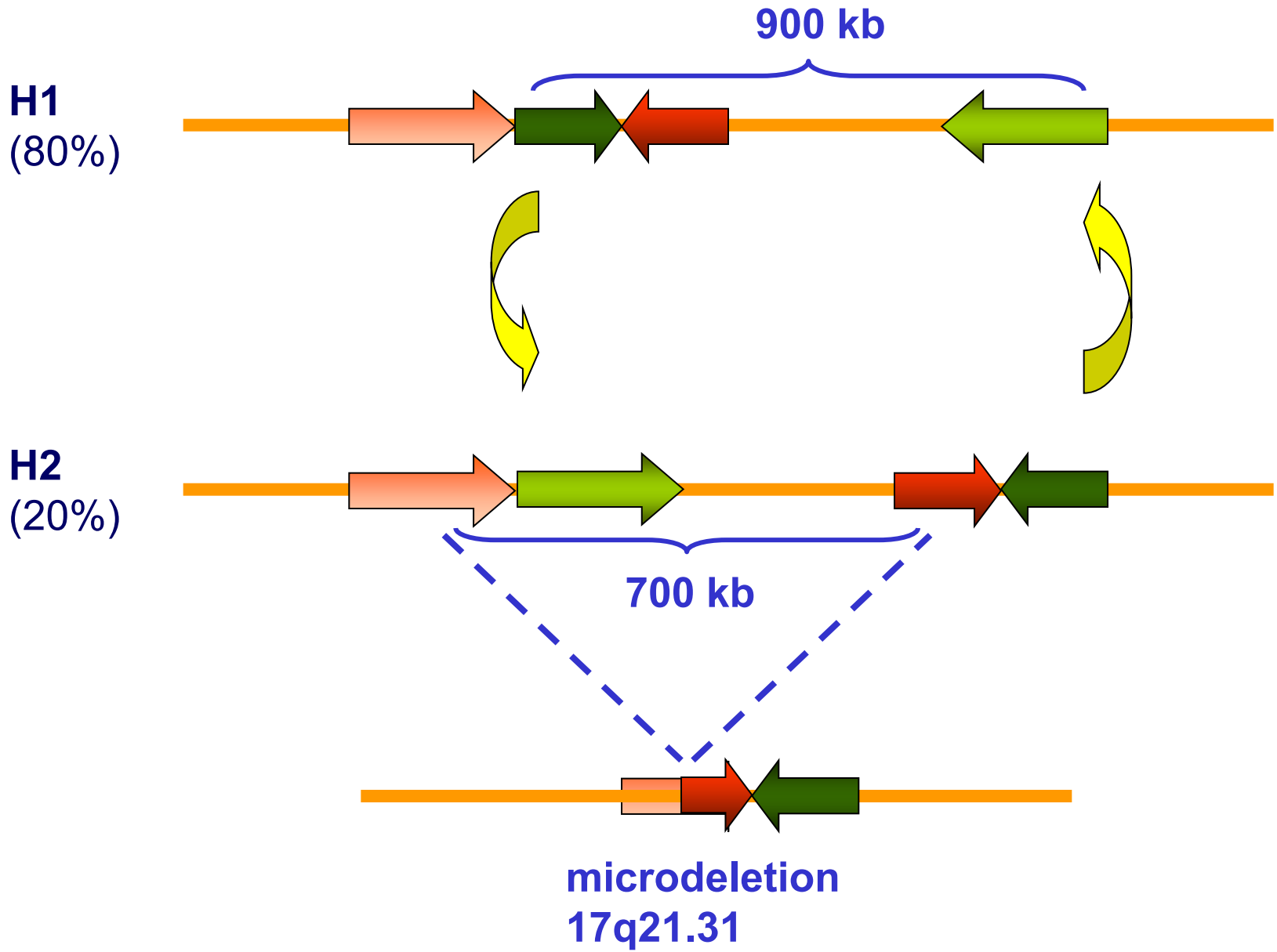
HNPP






22
22
1
23
500 kb
(HEM A)
F8 (Xq28)

Sporadic Mental Retardation (1% of cases)

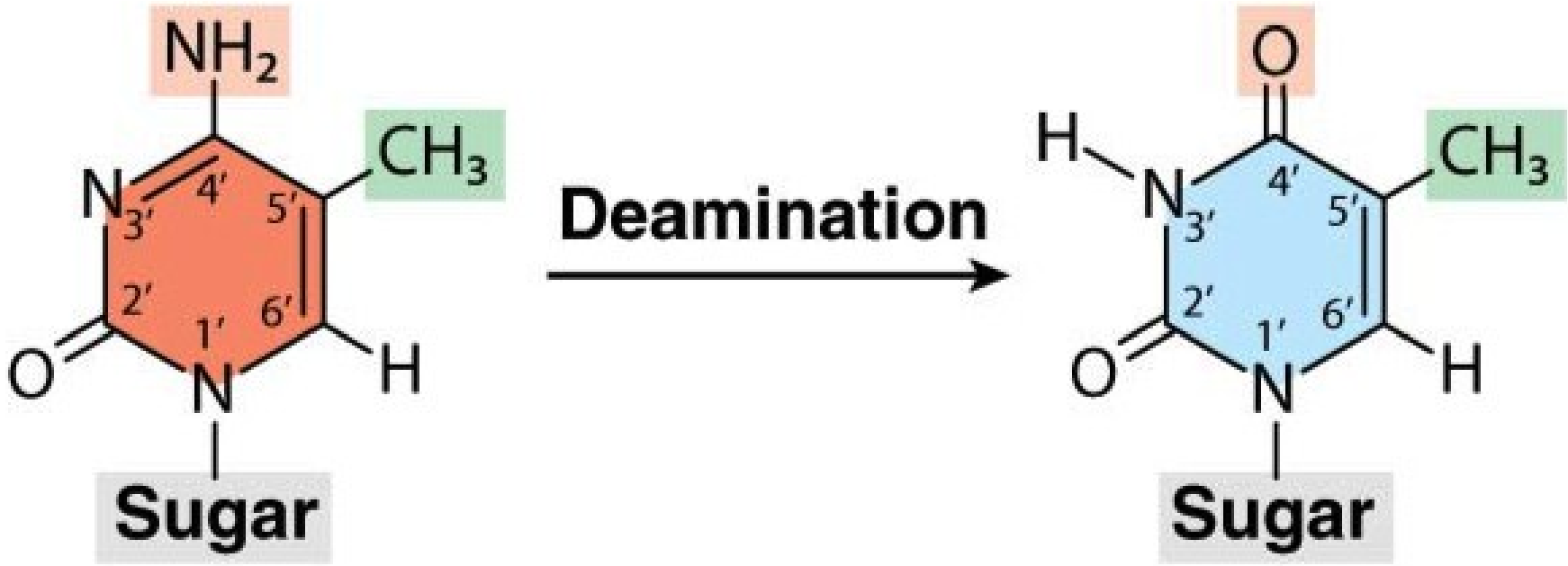


Lec 7.8. DNA damage and repair

Learning goals:

- How is DNA variation created? Mistakes happen.
- Mismatch Repair (MMR) is the mechanism that repairs loops and mismatches between normal nucleotides, arising from errors during replication
- Base Excision Repair (BER) and types of DNA lesions repaired by BER
- DNA lesions that distort the double helix are usually repaired by Nucleotide Excision Repair (NER). A similar repair pathway works during transcription
- Recombination not only creates new variants during meiosis, it also repairs Double-Strand Breaks (DSBs) during mitosis

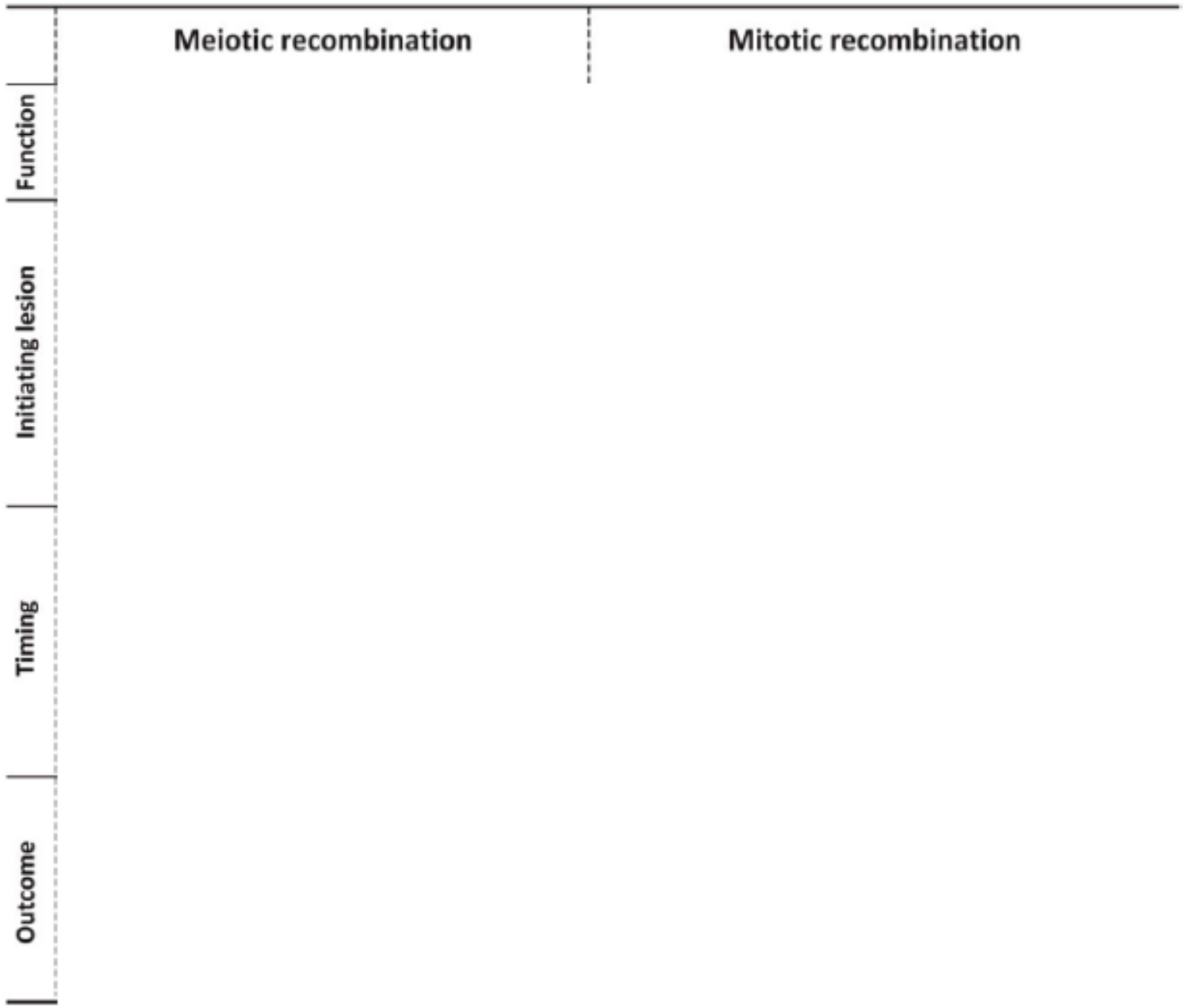
(b)

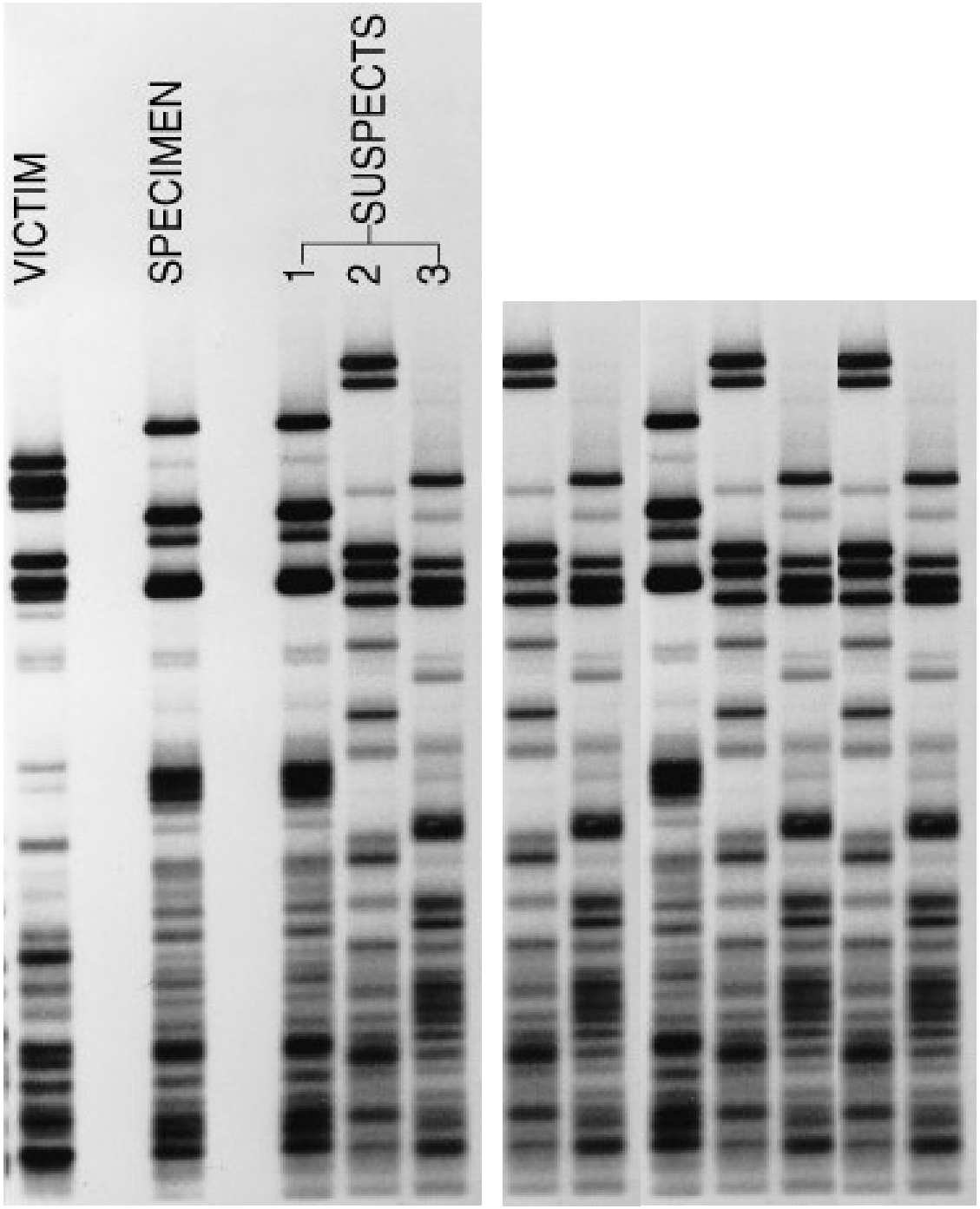


5-Methylcytosine (5-meC)

Thymine (T)

HR (meiosis) vs. **HRR** (mitosis)





Prosecutor's fallacy:

- Is #1 guilty?
- Is #1 innocent?
- Are #2 and #3 guilty?